

A Hybrid Cost Sensitive Ensemble Approach for Indonesian Local Script Handwritten Recognition

Bintang Nur Fadhlillah
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
bintang.fadhlillah@binus.ac.id

Dian Rakhmawati Lestari
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
dian.lestari002@binus.ac.id

Fadhlan Nur Rachman
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
fadhlan.rachman@binus.ac.id

Henry Lucky
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
henry.lucky@binus.ac.id

Rifqi Alfinnur Charisma
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
rifqi.charisma@binus.ac.id

Abstract—This electronic document is a “live” template and already defines the components of your paper [title, text, heads, etc.] in its style sheet. **CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.* (Abstract)

Keywords—Nusantara Scripts, Handwritten Character Recognition (HCR), Hybrid Learning, Class Imbalance, Ensemble Learning.

I. INTRODUCTION

Indonesia is a country with a rich and diverse culture, one aspect of which is reflected in the various regional scripts found across the archipelago, including the Balinese, Batak, Javanese, Jawi, Kawi, Lampung, Lontara, Pallawa, and Sundanese scripts. For thousands of years, these scripts have preserved history, literature, and knowledge while serving as the visual identity of the nation’s cultural diversity. However, active literacy in these scripts has declined drastically due to the dominance of digital communication using the Latin alphabet in both formal education and daily life; consequently, this heritage is at risk of extinction unless digital preservation efforts are undertaken immediately [1]. Handwritten Character Recognition (HCR) technology offers a solution for digitizing manuscripts, inscriptions, and archival documents written in the scripts of the Indonesian archipelago. However, developing a robust HCR system must address challenges such as the highly variable and complex writing styles among individuals, as well as the scarcity of data for most of these scripts [2][3].

Previous research has identified two main paradigms in character recognition. First, the classical machine learning approach uses feature extraction methods such as HOG and Hu Moments, combined with classifier models such as SVM, KNN, and Random Forest. This approach is computationally efficient and performs very well on small datasets. Then, experiments with models like XGBoost and LightGBM were able to match and outperform VGG19 on Javanese, Sundanese, and Balinese script datasets under limited data conditions [2]. Experiments with KNN and Random Forest were no less effective and demonstrated stable performance in recognizing Lontara script based on Hu Moment features. Second, deep learning approaches based on CNNs and Vision Transformers are capable of effectively learning feature representations from raw pixels. Individual benchmark results show excellent performance: a 12-layer DCNN with near-perfect accuracy on 120 Javanese script characters [5], a CNN with elastic distortion augmentations on Batak Toba script [6],

and transfer learning from a pretrained MobileNetV2 framework achieving near-perfect accuracy of 99.8% on Lampung script [7]. Lightweight architectures such as MobileNetV2 have proven effective in addressing data limitations while maintaining strong feature representations.

However, there is a significant gap in the literature. No study has yet conducted an experiment involving all nine Nusantara scripts together in a single standardized experiment, as most studies focus only on one or two scripts [1][9]. Furthermore, the severe class imbalance in the dataset has not been explicitly addressed using cost-sensitive learning strategies in the context of multi-script Nusantara HCR, so the evaluation metrics used tend to be biased toward the majority class [10]. Finally, no study has combined pretrained CNN feature extraction with a gradient-boosting-based ensemble classifier for an imbalanced multi-script setting; thus, the potential for conducting a DCNN feature extraction experiment with tree-based ensemble efficiency has not yet been systematically explored [2][9].

To fill this gap, this study proposes an experimental framework called Cost-Sensitive Hybrid Deep Feature Extraction and Ensemble Classification, which separates the feature representation stage from the classification stage. A lightweight pretrained CNN backbone from ImageNet, namely MobileNetV2 or EfficientNet-B0 [8][11], is used as a frozen feature extraction method that generates feature embeddings without the need for gradient-based retraining. These embeddings are then classified by three cost-sensitive gradient boosting ensemble models such as XGBoost, LightGBM, and CatBoost [12][13][14]. Class weights are calculated analytically from the class frequency distribution in the training set using the weight-balanced formula from Scikit-Learn, so that the model incurs a large penalty for misclassifying the minority class without using memory-intensive oversampling and undersampling techniques.

The framework developed offers distinct advantages over the two existing paradigms. Compared to classical ML, the use of pretrained CNNs for feature extraction can mitigate the risks associated with manual feature engineering, which is prone to human error and bias. Compared to deep learning alone, this hybrid architecture significantly reduces computational costs and data requirements, making it suitable for application even to minority scripts with limited data. The cost-sensitive weighting mechanism ensures a fair and balanced evaluation of all 228 character classes across 9 scripts. Beyond its technical contributions, this research also directly contributes to Sustainable Development Goal (SDG)

4: Quality Education through the development of digitalization tools that make the archipelago's heritage easily accessible to educators and students [15], as well as SDG 11: Sustainable Cities and Communities by providing computational infrastructure for cultural institutions to digitize their manuscript collections in a measured and scalable manner [16]. The open design of this research including the dataset, source code, and model weights, all of which are publicly available ensures that these benefits can be equally accessed by researchers who will continue this work.

II. RELATED WORK

A. Shallow Machine Learning for Script Recognition

Classic machine learning approaches have become a well-established baseline in handwritten character recognition, particularly under conditions of limited data. Sulistiyo (2025) evaluated XGBoost, LightGBM, and Random Forest against VGG19 on three Nusantara scripts (Javanese, Sundanese, Balinese) and found that shallow ML methods could match or outperform VGG19 under limited data conditions, confirming their relevance as computationally efficient alternatives [2]. Mardiani et al. (2025) confirmed the stable performance of KNN, Decision Tree, and Random Forest in Lontara script character recognition using Hu Moment features, demonstrating that geometry-based feature approaches remain relevant for scripts with limited morphology [4]. However, a fundamental limitation of this approach was identified by Handayani et al. (2022), who reported that KNN with projection profile segmentation achieved only 52% accuracy on high-texture Balinese Lontar manuscripts, far below the 92% achieved on clean paper samples—highlighting a critical dependence on preprocessing quality [17]. Firmansyah et al. (2025) also noted that Naive Bayes with zone-based feature extraction achieved only 65% accuracy on Nglegena Javanese script, revealing the limitations of traditional ML without complex feature engineering [3]. These findings underpin this study's decision to utilize CNNs as a more robust automatic feature extractor, rather than relying on manual feature descriptors.

B. Data Augmentation and Handling Class Imbalance

Data scarcity and class imbalance are the most persistent challenges in research on the recognition of Nusantara scripts. Cahyani et al. (2025) applied Enhanced Balancing GAN (EBGAN) to ancient Javanese script manuscripts and successfully improved the F1-score from 95.93% to 96.77% through the synthesis of minority class samples, demonstrating the effectiveness of generative approaches in addressing data imbalance [10]. Nugraha et al. (2025) demonstrated a 3.73% accuracy improvement through image fusion using Top-hat Transform, CLAHE, and DWT on Kaganga script, specifically to address the contrast degradation common in indigenous handwriting [18]. The NusaAksara study (2025) underscores the urgency of this issue: state-of-the-art large language models (LLMs) and vision-language models (VLMs) such as GPT-4o and Llama 3.2 achieve near-zero performance on eight Nusantara scripts, confirming that general-purpose models cannot replace domain-specific labeled data [19]. Zulkifli et al. (2024) demonstrated that augmenting the dataset with additional Arabic data improved the accuracy of ResNet34 to 96% in the recognition of cursive Jawi characters [20]. In contrast to these approaches, this study adopts cost-sensitive weighting as a strategy for addressing data imbalance that does not require

data volume manipulation or computationally intensive generative processes, making it more memory-efficient.

C. Ensemble Learning: XGBoost, LightGBM, and CatBoost

The gradient boosting ensemble method has proven to be a competitive alternative to SVMs and deep learning for multi-class classification tasks with structured data. XGBoost constructs a regularized additive model using second-order gradient statistics, which reduces overfitting in high-dimensional feature spaces while supporting parallel tree construction [12]. LightGBM employs leaf-wise tree growth with Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), making it up to 20 times faster than XGBoost while maintaining comparable accuracy [13]. CatBoost introduces ordered boosting to eliminate target leakage through permutation-based residual computation—a property that is particularly beneficial in multi-class classification problems with many categories [14]. The structural compatibility between CNN feature vectors, which have histogram-like distribution properties, and tree-based splitting criteria makes these three methods ideal choices as downstream classifiers in a hybrid CNN-ensemble pipeline [2].

D. Feature Extraction with Deep Learning CNNs

CNNs have become the dominant paradigm in handwritten character recognition by automatically learning hierarchical feature representations from raw pixels. For Nusantara scripts, CNN-based systems have achieved the best results: a 12-layer DCNN with seven types of geometric augmentations reached 99.65% accuracy on 120 classes of Javanese characters [5], while a CNN with elastic distortion augmentations achieved 98.36% validation accuracy on the Batak Toba script [6]. HCR-Net, which utilizes some layers of VGG16, achieved an 11% improvement over the state-of-the-art with a 34% reduction in trainable parameters, demonstrating the efficiency of partial transfer learning for scripts with limited data resources [21]. For lightweight deployment, MobileNetV2—which uses inverted residual blocks with linear bottlenecks across 3.4 million parameters—offers highly efficient feature extraction [8], while EfficientNet-B0 applies compound scaling to the network's depth, width, and resolution for an optimal accuracy-efficiency balance [11]. Saputra et al. (2023) confirmed that a pretrained MobileNetV2 achieves an accuracy of up to 0.998 on the Lampung script, supporting its suitability for mobile-based deployment [7]. In this study, both backbones are operated as frozen feature extractors—only forward passes are performed without retraining—thereby significantly reducing computational costs without sacrificing the quality of visual feature representations.

III. METHODOLOGY

A. Experiment Flow

This study follows a structured experimental workflow from start to finish, designed to run a feature extraction pipeline using deep learning and a classification pipeline using gradient boosting for the recognition of human characters in local Indonesian scripts. The workflow is divided into four main stages: (1) Data collection and preprocessing, (2) Modeling, Feature Extraction and evaluation, (3) Results and application with real data testing real time. Figure 1 provides a comprehensive overview of the designed experiment.

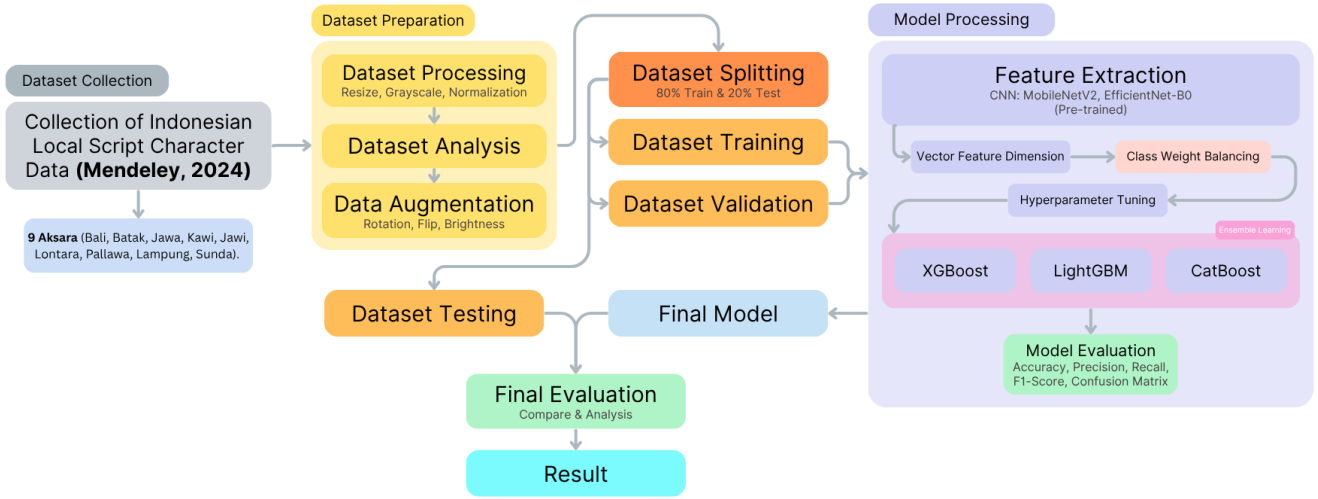


Fig 1. Flowchart Pipeline

The first phase, dataset preparation began with the collection of a data set of handwritten characters from the Indonesian local script, sourced from Mendeley Data (2024) [22]. This raw dataset was processed for standardization to next step, including binarization, pixel resizing, channel replication, normalization, exploratory data analysis, data augmentation, and splitting into training and validation sets with an 80:20 ratio for the subsequent model training phase.

The second is modelling and evaluation, where features must first be extracted using a frozen CNN since only the feature extraction layers are used, while the remaining layers (such as the classification layers) are removed because the next step involves calculating class weights and proceeding with gradient boosting classification training and hyperparameter tuning. The final step of this phase is evaluation using predefined metrics such as Macro F1, Precision, Recall, F1-Score, and the Confusion Matrix for each class

The third and final phase involves the submission of a set of test data that has been set aside and updated. The results of this phase constitute the final findings of this study, which will then be implemented via a web interface based on Streamlit or FastAPI for real-time HCR classification.

B. Dataset

The dataset used is the Indonesia Local Script Characters Handwriting Dataset, which contains images of handwritten characters from nine scripts of the Indonesian archipelago: Balinese, Batak, Javanese, Jawi, Kawi, Lampung, Lontara, Pallawa, and Sundanese [22]. This dataset contains 55,713 handwritten images and exhibits distinct characteristics in terms of variation, resolution, and class distribution, necessitating further analysis. Given the significant data imbalance between majority classes, such as the Sunda class, and minority classes, such as the Jawi class, single-class accuracy cannot be used as an evaluation metric because it risks introducing bias toward the majority class, as shown in Table 1.

Therefore, the data processing strategy includes the implementation of Cost-Sensitive Learning. During the classification modeling phase, class weight parameters will be calculated automatically using the `compute_class_weight` function from the Scikit-Learn library. This approach allows the model to assign a higher penalty to prediction errors for

minority classes such as Jawi and Batak, thereby improving the model's sensitivity without the need to manipulate data volume such as through oversampling which can burden computational memory.

TABLE I. DATASET COMPOSITION

Script	Region	Classes	Images	Percentage
Bali	Bali	18	4493	8.06%
Batak	NorthSumatera	19	1900	3.41%
Jawa	Central Java	20	9995	17.94%
Jawi	Malay Region	32	1600	2.87%
Kawi	Historical Java	31	6246	11.21%
Lampung	Lampung	20	4996	8.97%
Lontara	South Sulawesi	23	2299	4.13%
Pallawa	South Asia	33	3960	7.11%
Sunda	West Java	32	20224	36.30%
Total :		228	55713	100%

All images in the dataset are in JPEG or PNG format. Images vary in resolution and colour mode across scripts. During the preprocessing stage IIIA. all images are standardized to a uniform 224×224 pixel grayscale representation to ensure consistent input dimensions across all models. Next, the dataset was split using a stratified split method into an 80% training set and a 20% validation/test set. This ensures that each class is represented in an identical proportion in both evaluation phases, thereby preventing data distribution bias.

C. Feature Extraction

Unlike traditional machine learning approaches that rely on manual feature engineering, such as HOG or Hu Moments, this study implements an automatic feature extraction method using a Convolutional Neural Network (CNN) architecture. This approach was chosen to eliminate potential human error and bias in determining visual feature extraction.

This stage begins with image preprocessing, where Binarization and Thresholding techniques are applied to

separate character stroke features from background noise. Since pre-trained CNN architectures such as MobileNetV2 and EfficientNet-B0 require color (RGB) image input, the grayscale image is reconstructed through 3-channel RGB duplication and resized to 224×224 pixels [8] [11].

The processed images are then fed into a pre-trained ImageNet-based model. The final classification layer of the CNN architecture is removed. In this phase, the neural network acts purely as a frozen feature extractor. The model only performs a forward pass to map hierarchical visual representations such as lines, angles, and complex curves into a compact 1Dimensional Feature Vector, without undergoing retraining or backpropagation computations.

D. Model Experiment Setting

The 1D feature vectors extracted from the deep learning stage were used as independent variables to train three comparative ensemble learning models: XGBoost, LightGBM, and CatBoost [12][13][14]. To address the challenge of extreme class imbalance in the dataset, this study applied a cost-sensitive learning approach. Before the model training process begins, the penalty weights for each class are calculated analytically using the 'compute_class_weight' function from the Scikit-Learn library. This formula evaluates the frequency distribution of the images in the training data and automatically assigns larger penalty weights to the minority class. These analytical weights are then directly integrated into each model's objective function to prevent algorithmic bias toward the majority class, thereby making the model more sensitive without the need for data manipulation that could burden computational memory.

To achieve optimal classification performance, the hyperparameter optimization process was implemented by exploring a search space specific to each model architecture, as detailed in Table 2. The algorithm's learning capacity is controlled by adjusting the number of estimators within the range of 300 to 500 and the learning rate parameter between 0.05 and 0.2. To manage the complexity of the decision tree architecture, the XGBoost and CatBoost models were limited to a maximum depth of 6 to 10. Meanwhile, LightGBM was configured using a leaf-wise growth strategy with the number of leaves ranging from 31 to 127 to efficiently capture more complex feature patterns.

TABLE II. MODEL HYPERPARAMETER SETTINGS

Hyperparameter	XGBoost	LightGBM	CatBoost
n_estimators	300-500	300-500	300-500
max_depth	6-10	leaf-wise	6-10
learning_rate	0.05-0.2	0.05-0.2	0.05-0.2
subsample	0.7-1.0	0.7-1.0	-
colsample_bytree	0.7-1.0	0.7-1.0	-
Num_leaves	-	31-127	-
l2_leaf_reg	-	-	1-10
reg_alpha/lambda	0-1	0-1	-
random_states	42	42	42
eval_metric	mlogloss	multi_logloss	MultiClass

In addition to adjusting basic parameters, regularization techniques are also extensively applied to prevent the model from overfitting to the training data. This strategy includes adjusting row and column subsampling parameters within the range of 0.7 to 1.0, as well as applying L1/L2 penalty parameters configured according to each model's built-in mechanism. Given that this classification involves nine character targets, the loss function is specifically configured for a multiclass approach such as mlogloss and multi_logloss. To ensure that all experiments can be consistently reproduced, the random state is fixed at 42 in all stages. The model with the best configuration combination is then selected based on the highest F1-Score performance on the validation dataset.

E. Evaluation Metrics

Due to class imbalance, we need to evaluate model performance using four complementary metrics: precision, recall, and F1 score per class, with the macro F1 score as the primary comparison criterion. While accuracy metrics are intuitive, they can be misleading in conditions of class imbalance, as high accuracy can be achieved simply by overpredicting the majority class. Precision and Recall address this by focusing on per class prediction quality:

$$Precision = \frac{TP}{TP+FP}, \quad Recall = \frac{TP}{TP+FN} \quad (1)$$

Where Represents True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). The F1 score for that class balances precision and recall:

$$F1Score = 2 \frac{Precision \times Recall}{Precision+Recall} \quad (2)$$

Macro-F1 aggregates the per-class F1-scores by computing their unweighted mean across all K emotion classes:

$$MacroF1 = \frac{1}{K} \sum_{k=1}^K F1Score_k \quad (2)$$

This formulation assigns equal importance to each class regardless of sample count, making Macro-F1 particularly appropriate for imbalanced script recognition datasets where minority classes such as Jawi and Batak must not be systematically overlooked [23]. An example classification report illustrating these metrics is shown in Table III.

TABLE III. EXAMPLE OF CLASSIFICATION REPORT AND MACRO AVG

	Precision	Recall	F1Score	Support
Label 1	0.67	0.67	0.67	3
Label 2	0.25	1.00	0.40	1
Label 3	1.00	0.50	0.67	6
Accuracy			0.60	10
Macro Avg	0.64	0.72	0.58	10

IV. EXPERIMENT RESULTS

This is the template of the experiment results starts

V. CONCLUSION

This is the template of the conclusion starts

OPEN DATA

The dataset used in this study is publicly available. Dataset: Indonesian Local Script Characters Handwriting Dataset, published by A. F. Ihsan (2024) on Mendeley Data. Access: <https://data.mendeley.com/datasets/vfj32bpjsf/1>. The experimental code and configuration files will be made available in the supplementary repository upon paper acceptance.

AUTHOR CONTRIBUTION

Bintang Nur Fadhlillah: conceptualization of the research topic and terminology, identification of experimental variables and intervention design, and writing the introduction and model experiment settings. Dian Rakhmawati Lestari: conducting the systematic literature review, drafting the related work section, and performing manuscript formatting, review, and editing. Fadhlhan Nur Rachman: problem formulation and ideation, methodology design, development of the experimental pipeline and validation strategy, data visualization, and writing the introduction. All authors reviewed and approved the final manuscript. Henry Lucky: co-supervision, methodology validation, and writing – review and editing. All authors reviewed and approved the final manuscript. Rifqi Alfinnur Charisma: supervision, project administration, validation, and critical review of the manuscript.

REFERENCES

- [1] B. Krisna et al., "Deep Learning Approaches for Nusantara Scripts Optical Character Recognition," IJCCS (Indonesian Journal of Computing and Cybernetics Systems), 2023. [Online]. Available: <https://journal.ugm.ac.id/ijccs/article/view/86302/36989>
- [2] M. D. Sulistiyo, "Investigating Shallow Learning Methods for Optical Character Recognition of Indonesia's Nusantara Scripts", *J. RESTI (Rekayasa Sist. Teknol. Inf.)*, vol. 9, no. 6, pp. 1538 - 1550, Jan. 2025. Available: <https://jurnal.iaii.or.id/index.php/RESTI/article/view/6648/1200>
- [3] F. Firmansyah et al., "Optimization of Nglegena Javanese Script Recognition With Machine Learning Based on Zoning," IJODAS, 2025. [Online]. Available: <https://jurnal.yoctobrain.org/index.php/ijodas/article/view/256>
- [4] A. Mardiani et al., "Classification of Lontara Script Using K-NN Algorithm, Decision Tree, and Random Forest Based on Hu Moments and Canny Segmentation," IJODAS, 2025. [Online]. Available: <https://jurnal.yoctobrain.org/index.php/ijodas/article/view/281>
- [5] W. Kusuma et al., "Handwritten Javanese Script Recognition Based on 12-Layers DCNN," IJAI, 2023. [Online]. Available: <https://ijai.iaescore.com/index.php/IJAI/article/view/22593/13718>
- [6] Rajagukguk et al., "Handwritten Batak Toba Script Recognition Based on Deep Learning Using CNN," J. AIEA, 2024. [Online]. Available: <https://ioinformatic.org/index.php/JAIEA/article/view/1795>
- [7] T. Saputra et al., "Development of Lampung Script Characters Recognition Model using TensorFlow," IJECS, 2023. [Online]. Available: <https://ejournal.radenintan.ac.id/index.php/IJECS/article/view/19878/pdf>
- [8] M. Sandler et al., "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in Proc. CVPR, 2018. [Online]. Available: <https://arxiv.org/abs/1801.04381>
- [9] D. Gunawan et al., "Systematic Literature Review Deep Learning pada Pengenalan Aksara Jawa untuk MobileNetV3-SVM," Proseminas LPPM UNISA Yogyakarta, 2026. [Online]. Available: <https://proceeding.unisavogya.ac.id/index.php/proseminaslppm/article/view/2026>
- [10] R. Cahyani et al., "Evaluation of Synthetic Data Effectiveness Using GAN in Improving Javanese Script Recognition on Ancient Manuscripts," JUTI ITS, 2025. [Online]. Available: <https://juti.if.its.ac.id/index.php/juti/article/view/1256>
- [11] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in Proc. ICML, Long Beach, CA, USA, 2019. [Online]. Available: <https://arxiv.org/abs/1905.11946>
- [12] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining (KDD), San Francisco, CA, USA, 2016, pp. 785–794. [Online]. Available: <https://dl.acm.org/doi/10.1145/2939672.2939785>
- [13] G. Ke et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2017. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>
- [14] L. Prokhorenkova et al., "CatBoost: Unbiased Boosting with Categorical Features," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2018. [Online]. Available: <https://arxiv.org/abs/1706.09516>
- [15] United Nations, "Goal 4: Quality Education," Sustainable Development Goals, 2015. [Online]. Available: <https://sdgs.un.org/goals/goal4>
- [16] United Nations, "Goal 11: Sustainable Cities and Communities," Sustainable Development Goals, 2015. [Online]. Available: <https://sdgs.un.org/goals/goal11>
- [17] I. K. Handayani et al., "Handwritten Balinese Script Recognition Using Projection Profile and KNN," Semantic Scholar, 2022. [Online]. Available: <https://pdfs.semanticscholar.org/2925/45eabcd92304b053a0d68f1021ab5f39bce4.pdf>
- [18] B. Nugraha et al., "Handwritten Kaganga Script Classification Using Deep Learning and Image Fusion," ResearchGate, 2025. [Online]. Available: <https://www.researchgate.net/publication/390377012>
- [19] A. Santoso et al., "NusaAksara: A Multimodal and Multilingual Benchmark for Preserving Indonesian Indigenous Scripts," in Proc. ACL, 2025. [Online]. Available: <https://aclanthology.org/2025.acl-long.1377.pdf>
- [20] M. Zulkifli et al., "Augmentation of Additional Arabic Dataset for Jawi Writing and Classification Using Deep Learning," J. Rekayasa Elektrika, 2024. [Online]. Available: <https://jurnal.usk.ac.id/JRE/article/view/33722>
- [21] S. Fauzi et al., "HCR-Net: A Deep Learning Based Script Independent Handwritten Character Recognition Network," arXiv, 2024. [Online]. Available: <https://arxiv.org/pdf/2108.06663>
- [22] F. Ihsan, "Indonesian Local Script Characters Handwriting Dataset," Mendeley Data, V1, 2024. Tersedia: <https://data.mendeley.com/datasets/vfj32bpjsf/1>. doi: 10.17632/vfj32bpjsf.1.
- [23] Leung K. Micro, macro & weighted averages of F1 score, clearly explained. Towards Data Science; 2022. Available from: <https://towardsdatascience.com/micro-macro-weighted-averages-of-f1-score-clearly-explained-b603420b292f/>.